W04 报告

1. 测试范围与环境

本文核对本地向量库、三路召回、端到端探针，以及五个 Dart 包的默认单测覆盖率。默认套件排除 `integration`，不加载模型和数据集。冻结的 25 条证据集没有改。

环境见 `logs/env_w4.txt`：

- Dart 3.13.4（stable）
- Java 17.0.4
- Python 3.13.2（`engine/.venv`）
- `C:\Windows\System32\winsqlite3.dll` 存在。容量探针读到 SQLite 3.53.4，WAL
- chromadb 1.5.9，见 `logs/chroma_baseline.txt`
- `datasets/samples/sample.pdf`、`sample.docx`、`sample.png` 和 COCO 披萨图都在

判定以日志里的退出码、计数和耗时为准。终端截屏在第 5 节，路径相对本报告。若干 `e2e_*.txt` 的 `query=` 行是 PowerShell 把 Dart 的 UTF-8 输出按系统代码页解开的，命令注释行仍是原来的中文问句。

机器网卡没有关掉。离线核对是三个 `lib/` 的 grep，加上一次只拦 Dart 和 Python 出站的防火墙规则。规则已经删除。见 OFF-01。

2. 用例表

用例 ID	命令	期望	实际	判定	证据
ANA-01	三个新包各自 `dart analyze`	0 issue	三个包都是 `No issues found!`，退出码 0	通过	`logs/dart_analyze.txt`；`screenshots/ANA-01-dart-analyze.png`
UNIT-01	三包 `dart test --exclude-tags integration`	默认套件全绿	2026-10-26：存储 `00:02 +23`，检索 `00:02 +9`，编排 `00:03 +23`，见 `logs/coverage.txt`。2026-10-27 只重跑存储和检索：`00:03 +24`、`00:02 +11`。编排没有重跑	通过	`logs/coverage.txt`；`logs/coverage_20261027_storage_retrieval.txt`

用例 ID	命令	期望	实际	判定	证据
COV-01	五包 `--coverage` 后 `--report-on=lib`	新三包 ≥ 85%；解析不低于 80%；嵌入不低于 85%	存储 574/666 = 86.19%，检索 158/164 = 96.34%，编排 272/298 = 91.28%，解析 979/1091 = 89.73%，嵌入 1250/1458 = 85.73%	通过	覆盖率报告.md；`coverage/*.lcov.info`；`screenshots/COV-01-coverage.png`
H1	Chroma 与 Dart Top-5 对照	125 个位置一致率 ≥ 96%	`agreed=125, agreement=1.000000, passed=1`	通过	`logs/storage_equivalence.txt`；`screenshots/H1-storage-equivalence.png`
CAP-01	`packages/core_storage/bin/capacity_probe.dart` 写 1 万条	≤ 600 ms	`write_ms=444`。SQLite 3.53.4，WAL	通过	`logs/storage_capacity.txt`；`screenshots/CAP-02-capacity.png`
CAP-02	同上，384 维查询中位	≤ 60 ms	中位 44 ms，最小 41，最大 50。`EXPLAIN` 走 `idx_chunk_space`	通过	`logs/storage_capacity.txt`；`screenshots/CAP-02-capacity.png`
CHR-01	`engine/chroma_baseline.py`	记下 chromadb 版本、Hit@5、MRR@10。这是导出向量上的对照，不是产品检索器	chromadb 1.5.9，`hnsw:space=cosine`。Hit@5 0.880000 [0.700442, 0.958332] (22/25)，MRR@10 0.795714。对 numpy 125/125	通过	`logs/chroma_baseline.txt`；`screenshots/CHR-01-chroma-baseline.png`
E2E-01	`index_probe` 建 `reports/week4/.index/evidence.sqlite`	样本和 COCO 小样入库	样本 3/3，97 块，12045 ms。COCO 50/50，50 块，7048 ms。合计 147 块、53 个文档，`index_ms_total=19093`。`keyword_terms=0`	通过	`logs/e2e_build.txt`；`logs/e2e_samples.txt`；`screenshots/E2E-01-build.png`
E2E-02	同一库再跑一次 `index_probe`	内容未改则全部复用	样本 `reused=3`，COCO `reused=50, written=0`。统计仍是 147/53。`index_ms_total=6873`	通过	`logs/e2e_rebuild.txt`；`screenshots/E2E-02-rebuild.png`

用例 ID	命令	期望	实际	判定	证据
E2E-03	临时目录三份 Markdown，改 `a.md` 后再索引	未改的文件复用；改过的文件替换旧行，文档数不增加	第一次 3 个文档、378 块、16130 ms。追加一句后 `written=1, reused=2,` 85 块，5117 ms。`stats_documents=3, stats_chunks=379`	通过	`logs/e2e_incremental.txt`；`screenshots/E2E-03-incremental.png`
E2E-04 / H3	`search_probe --top 10 --query "工具靠什么判断文件类型？"`	问句不含文件名。Top-5 出现目标文件，并给出页和偏移。原文件在磁盘上	Top-10 都是文本。`sample.pdf` 在第 1、2、4、5 名，`sample.docx` 在第 3 名。第 4 名 `page=1, start=617, end=710`，片段含 `missing or misleading extension`（证据跨度 679–710 落在这块内）。两个目标文件都存在。这次记录是权重 0.2。2026-10-27 用 0.49 重测同一句：`sample.pdf` 仍在第 1、2 名，第 3 名是 `sample.png`（`clipShared`），第 5 名是 `000000463730.jpg, 617–710` 落到**第 6 名**，Top-5 里不再有这句。H3 的判据是「Top-5 出现目标文件并给出页和偏移」，第 1 名的 `423–517` 仍满足，判定不变	通过	`logs/e2e_query.txt`；`logs/e2e04_rank_after_reweight_20261027.txt`；`logs/env_w4.txt`；`screenshots/E2E-04-h3-query.png`
E2E-05	同一进程第二条问句「这份说明在能够被完全信任之前，还要求人做什么？」	再出一份带出处的 Top-K	第 1 名 `sample.pdf, page=1, start=160, end=253`，片段含 `requires human review`。`encode_ms=67`	通过	`logs/e2e_query.txt`；`screenshots/E2E-05-second-query.png`

用例 ID	命令	期望	实际	判定	证据
E2E-06	`--extension .pdf`、`--path-prefix sample.pdf`、`--modified-after`	扩展名和路径只留 pdf；过去的时间仍有命中；2099 年之后为 0	`.pdf` 与 `sample.pdf` 的前几名都是 `sample.pdf`。`2020-01-01` 的第 3 名仍是 `sample.docx`。`2099-01-01` 为 `hits=0`	通过	`logs/e2e_filters.txt`；`screenshots/E2E-06-filters.png`
E2E-07	问「哪张图里有人在厨房做披萨？」，以及 `--extension .jpg`	Top-5 出现 `000000397133.jpg`	不限类型时 Top-10 全是 `sample.pdf` / `sample.docx`。只留 `.jpg` 时五条都是图，第 1 名是 `000000403385.jpg`，披萨图不在这五条里。英文证据句只走 clipShared 时该图排第 1。2026-10-27 用 0.49 重测本行判据：不限类型时 `000000397133.jpg` 在第 3 名，加 `--extension .jpg` 时第 1 名就是它，Top-5 出现目标文件这一条已满足。本行仍按 2026-10-25 记录判「未通过」；重测见 `logs/e2e07_rerun_20261027.txt`，不改这一行	未通过	`logs/e2e_query.txt`；`logs/e2e_filters.txt`；`logs/e2e07_clip_diagnosis.txt`；`logs/e2e07_rerun_20261027.txt`；`screenshots/E2E-07-image.png`
E2E-08	同一问句比较 `--arm A5` 与 `--arm A4`	邻域不改名次和偏移；A5 的片段带上相邻块	两边第 4 名都是 `score=0.007813, page=1, start=617, end=710`。A5 片段多出前一块的 `interface, making Tika useful`。这次记录是权重 0.2，命令是 `--top 5`。2026-10-27 用 0.49 重测同一句的 A4 与 A5：两边第 4 名都是 `sample.docx`（306–426），第 5 名都是 `000000463730.jpg`，617–710 都在第 6 名，仍不进 `--top 5` 的输出。A5 与 A4	通过	`logs/e2e_neighborhood.txt`；`logs/e2e04_rank_after_reweight_20261027.txt`；`screenshots/E2E-08-neighborhood.png`

用例 ID	命令	期望	实际	判定	证据
			在这句上名次和偏移仍相同，邻域不改名次这条成立；但被比较的第 4 名已不是 617–710，这条用例的比对对象随权重一起变了		
E2E-09	index_probe --cancel-after 1，目录里 a.txt、b.txt、c.txt	已写入的文件留下，后面的文件不写	written=1, cancelled=true, stats_documents=1。检索命中 a.txt。问第二句仍返回这一行，因为库里只有一行，且没有分数下限	通过	logs/e2e_cancel.txt; screenshots/E2E-09-cancel.png
E2E-10	目录里一份好 txt 和一份坏 docx	坏文件记失败，好文件仍入库，整批退出码 0	ingest_failed=1, written=1，报告 failed=0。检索命中 good.txt	通过	logs/e2e_cancel.txt; screenshots/E2E-10-bad-file.png
E2E-11	空库上 search_probe	hits=0	清库前的空库查询 hits=0	通过	logs/e2e_clear.txt; screenshots/E2E-11-empty.png
E2E-12	写入后再 --clear，然后查询	块数从 1 到 0，查询为空	clear_before_chunks=1, clear_after_chunks=0，随后 hits=0	通过	logs/e2e_clear.txt; screenshots/E2E-12-clear.png
DOC-01	index_probe --path docs 两次，再搜「产品侧用系统 SQLite 保存向量」	第一次写入，第二次复用。中文问句能命中 ADR	9 个文件、2870 块，第一次 110092 ms，第二次 reused=9，5481 ms。keyword_terms=14915。第 1 名是 ADR 0004，start=1304，end=1336，通道 documentText,keyword	通过	logs/e2e_docs.txt
H2	eval_retrieval.py 六臂 dev，然后 A4 holdout 一次	融合的 Hit@5 与 MRR@10 不低于任一单通道。没有增益就照实写，不改权重	dev 上 A4 与 A1 同为 10/15、MRR 0.622222，高于 A0 和 A2。没有正增益。holdout A4 为 5/10。parameters_changed_after_freeze=0，holdout 只打了 1 次	通过（无正增益）	检索精度基准报告.md; logs/retrieval_bench_dev.txt; logs/retrieval_bench_holdout.txt; screenshots/H2-dev.png;

用例 ID	命令	期望	实际	判定	证据
					screenshots/H2-holdout.png
OFF-01	三个新包 lib/ 搜索 HttpClient|Socket|import .http|package:http，并在出站被拦时对已建好的库做查询	无匹配；外部地址失败时，本地查询仍有命中；规则事后不留	扫描 40 个 dart 文件，matches=0。unshare -rn 里网络与 DNS 不可达，索引与语料可读。防火墙：加规则前 223.5.5.5:443 能连；规则生效时 Python 为 WinError 10013，Dart 出站失败。同时 search_probe --arm A5 问「工具靠什么判断文件类型？」search_exit=0，第 1 名 sample.pdf，第 4 名 start=617 end=710。删规则后再 show 为 No rules match，该地址恢复。第 4 名是权重 0.2 的记录。2026-10-27 用 0.49 重测同一句：617–710 在第 6 名，第 1 名仍是 sample.pdf（423–517）。断网判据取的是「本地查询仍有命中」，不依赖第 4 名。防火墙那一段没有重跑，权重与规则无关	通过	logs/offline_check.txt；logs/offline_namespace.txt；logs/offline_firewall.txt；logs/e2e04_rank_after_reweight_20261027.txt；screenshots/OFF-01-offline-grep.png
NEI-01	engine/eval_neighborhood.py 用 evidence_set_w4.jsonl 验证邻域扩展	12 条题都能对齐块；跨块题能被 ±1 邻域覆盖	chunk_coverage_ok=12/12，cross_chunk_rows=7。严格口径 reachable_with_one_hop=5（forall，即原来的 5/7）。另计 exists_one_anchor=6：跨度 2 的五条两口径都过；w4-04 跨度 3 只过	部分通过	logs/e2e_neighborhood_w4.txt；证据集说明_w4.md

用例 ID	命令	期望	实际	判定	证据
			exists；w4-03 跨度 4 两口径都不过		
QRY-01	`search_probe --top 5`，先 `--arm A2` 再 `--arm A4`，五条图像问句	文档文本塔用原问句。词表命中时 CLIP 文本塔用离线英文词。有中文 bigram 时，长度不少于 3 的拉丁串和英文词再走 trigram。不并入冻结的 25 条分数，也不改 E2E-07	2026-10-27，权重仍是 `(0.5, 0.3, 0.2)`。五条 A2 的金标都进 Top-5：披萨、自行车、客机和 `sample.png` 是第 1，浴室是第 2。四条场景问句 `keyword_hits=0`。A4 的 Top-5 仍是 `sample.pdf` / `sample.docx`。`sample.png` 的 A2 第 1 只说明文件被召回	通过	`logs/query_rewrite_probe.txt`；检索精度基准报告.md 第 9 节

E2E-01 到 E2E-12 里，E2E-07 未通过，其余通过。NEI-01 的严格口径仍是 5/7，w4-03 与 w4-04 的 forall 未覆盖；存在锚点口径是 6/7。

2026-10-27 先改了查询，再把图像权重从 0.2 改为 0.49。上面的用例表和六臂表仍是权重 0.2 的记录，E2E-07 那一行保持未通过。新测量在 `logs/retrieval_bench_reweight_dev.txt`。同一句披萨问句的融合名次是 3，`000000397133.jpg` 在 Top-5 里。蓝自行车、破损淋浴墙、客机也在 Top-5 里。版面标题那条仍不在，Top-5 还是两份说明。文本和元数据十条仍全部命中。holdout 的 9/10 写在精度报告第 10 节，是第二次查看，不改上面 H2 的 5/10，也不能与 0.500 并列成提升。

3. 交付物

编号	文件	状态
G1	`packages/core_storage/`、`packages/core_retrieval/`	已完成。分析无问题，默认套件全绿。COV-01 仍是 2026-10-26 的 86.19% 与 96.34%。2026-10-27 重跑这两包：存储 637/729 = 87.38%，检索 159/165 = 96.36%
G2	`packages/core_pipeline/bin/index_probe.dart`、`search_probe.dart`	已完成。H3 通过，见 E2E-04
G3	本报告、`reports/week4/logs/`、`screenshots/`	已完成。E2E-07 未通过，其余用例通过。20 张终端截屏
G4	`reports/week4/检索精度基准报告.md`	已完成。六臂、Wilson 区间、逐题表。第 2 节的 holdout 仍是一次。第 10 节的 0.900 是第二次查看
G5	`reports/week4/logs/storage_equivalence.txt`	已完成。125/125，脚本与原始 jsonl 已对齐
G6	`docs/architecture/adr/0004-vector-storage-and-fusion.md`	已完成。决策 1–7 各含 Alternatives/Consequences，决策 8 以「运行时」小节挂在决策 1 下并补齐这两节；

编号	文件	状态
		融合后果写明不低于单通道且无正增益
G7	docs/api/storage-module-api.md、docs/api/retrieval-module-api.md	已完成。存储文档写明 1 万块写入 444 ms、查询中位 44 ms。代次判等是 6 个字段，即决策 4 的五项，解析器名和版本分计
G8	reports/week4/coverage/*.lcov.info、覆盖率报告.md	已完成。五个包在 2026-10-26 都在门槛之上。2026-10-27 的两包副本另档，不覆盖当天的 lcov

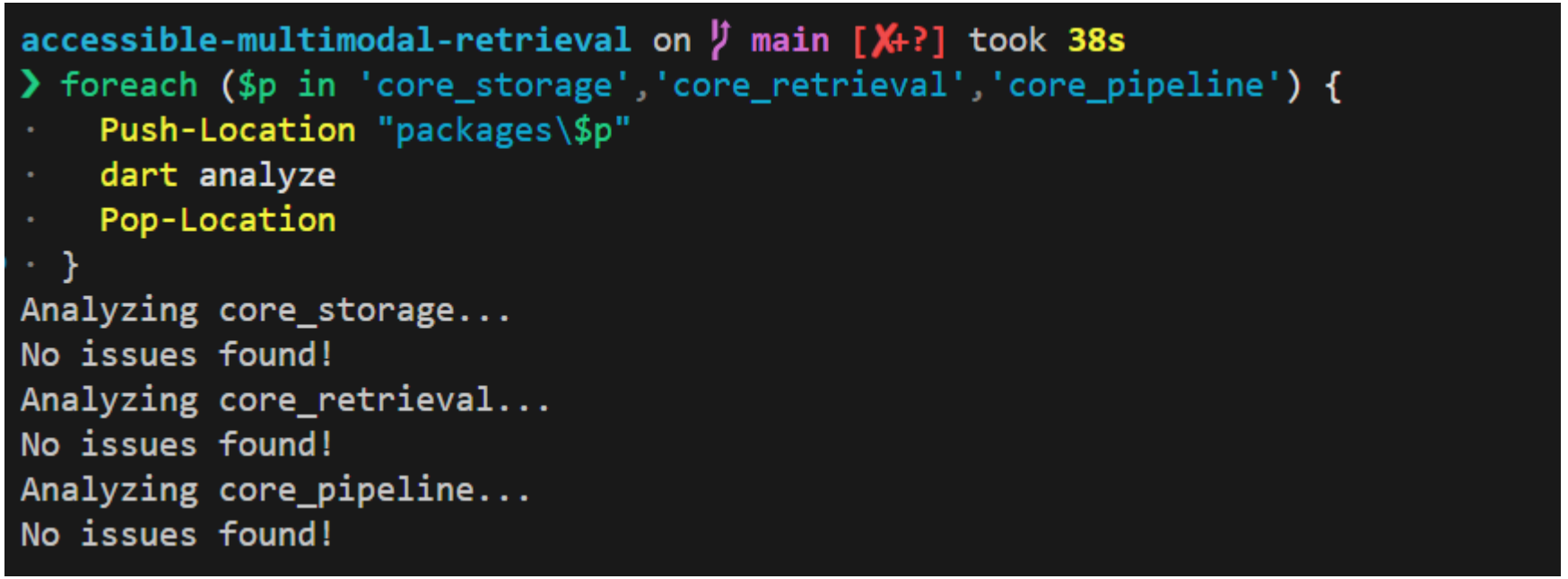
reports/week4/evidence_set_w4.jsonl 有 12 条，说明在 证据集说明_w4.md。它没有并入冻结集，也没有参加打分；本轮由 engine/eval_neighborhood.py 首次用于邻域验证，结果见 NEI-01。

4. 证据索引

路径	对应内容
reports/week4/logs/dart_analyze.txt	ANA-01
reports/week4/logs/coverage.txt	UNIT-01、COV-01 的合计。2026-10-26
reports/week4/logs/coverage_20261027_storage_retrieval.txt	存储与检索的后一次覆盖率。不改 COV-01
reports/week4/coverage/*.lcov.info	COV-01 的五份，加上 2026-10-27 的 core_storage_20261027.lcov.info 与 core_retrieval_20261027.lcov.info
reports/week4/logs/storage_equivalence.txt	H1
reports/week4/logs/storage_capacity.txt	CAP-01、CAP-02
reports/week4/logs/chroma_baseline.txt	CHR-01
reports/week4/logs/e2e_build.txt	E2E-01
reports/week4/logs/e2e_samples.txt	E2E-01 的副本
reports/week4/logs/e2e_rebuild.txt	E2E-02
reports/week4/logs/e2e_incremental.txt	E2E-03
reports/week4/logs/e2e_query.txt	E2E-04、E2E-05、E2E-07
reports/week4/logs/e2e07_clip_diagnosis.txt	E2E-07 的三条诊断。中文问句仍未通过，不并进 25 条分数
reports/week4/logs/e2e_filters.txt	E2E-06、E2E-07 的 jpg 过滤
reports/week4/logs/e2e_neighborhood.txt	E2E-08
reports/week4/logs/e2e04_rank_after_reweight_20261027.txt	E2E-04、E2E-08、OFF-01 在权重 0.49 下的重测。617–710 第 6 名
reports/week4/logs/e2e07_rerun_20261027.txt	E2E-07 在权重 0.49 下的重测。判据已满足，本行仍判未通过

路径	对应内容
reports/week4/logs/e2e_cancel.txt	E2E-09、E2E-10
reports/week4/logs/e2e_clear.txt	E2E-11、E2E-12
reports/week4/logs/e2e_docs.txt	DOC-01
reports/week4/logs/retrieval_bench_dev.txt	H2 的 dev
reports/week4/logs/retrieval_bench_holdout.txt	H2 的 holdout
reports/week4/logs/offline_check.txt	OFF-01 的 grep 门禁与断网说明
reports/week4/logs/offline_namespace.txt	OFF-01 的断网命名空间取证（无路由、无 DNS）
reports/week4/logs/offline_firewall.txt	OFF-01 的出站规则。查询在规则生效时命中，规则已删除
reports/week4/logs/e2e_neighborhood_w4.txt	NEI-01 的邻域验证
reports/week4/logs/keyword_path_probe.txt	关键词两条路径。不并入冻结的 25 条分数
reports/week4/logs/query_rewrite_probe.txt	QRY-01。2026-10-27 查询改写，当时图像权重仍是 0.2
reports/week4/logs/retrieval_bench_reweight_dev.txt	图像权重 0.49 的 dev。不替换冻结的六臂表
reports/week4/logs/retrieval_bench_reweight_holdout.txt	图像权重 0.49 的 holdout。第二次查看，0.900 不再是无偏估计，不替换第一次的 0.500
reports/week4/logs/env_w4.txt	环境与目标文件是否存在
reports/week4/检索精度基准报告.md	G4
reports/week4/覆盖率报告.md	G8
reports/week4/evidence_set_w4.jsonl	邻域核对，未打分
screenshots/*.png	第 2 节有截屏的用例

5. 截屏

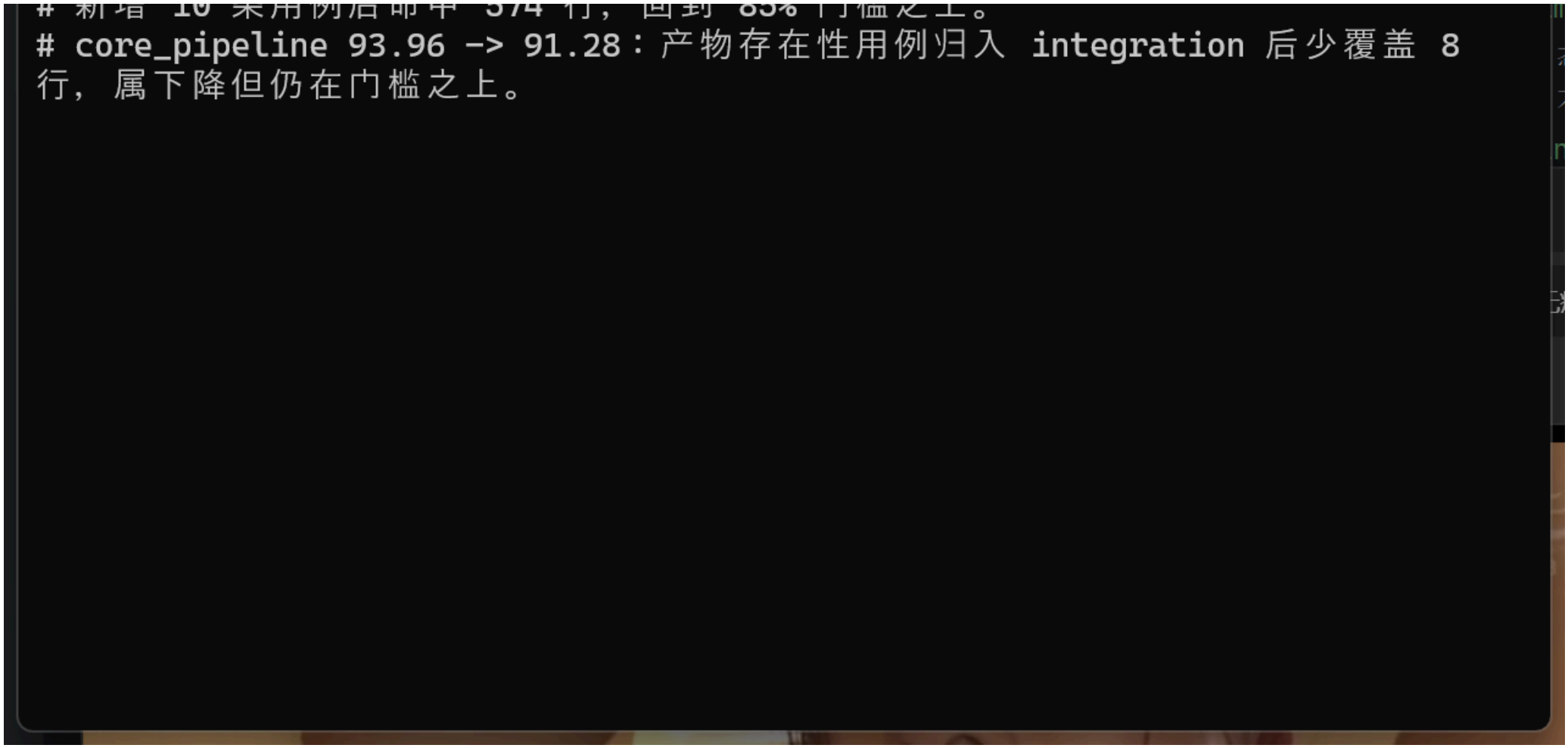

accessible-multimodal-retrieval on main [✗+?] took 38s
❯ foreach ($p in 'core_storage','core_retrieval','core_pipeline') {
· Push-Location "packages\$p"
· dart analyze
· Pop-Location
· }
Analyzing core_storage...
No issues found!
Analyzing core_retrieval...
No issues found!
Analyzing core_pipeline...
No issues found!

W4SHOT-COV

```
PS E:\Project\accessible-multimodal-retrieval> Get-Content -LiteralPath rep
orts\week4\logs\coverage.txt
date=2026-10-26
# From each package directory. integration tests are excluded and do not lo
ad models.
#   dart test --coverage=coverage --exclude-tags integration
#   dart run coverage:format_coverage --lcov --in=coverage --out=coverage\l
cov.info --report-on=lib --package=.
# Copies are reports/week4/coverage/<package>.lcov.info

core_storage lines_hit=574 lines_found=666 pct=86.19 suite=00:02 +23
core_retrieval lines_hit=158 lines_found=164 pct=96.34 suite=00:02 +9
core_pipeline lines_hit=272 lines_found=298 pct=91.28 suite=00:03 +23
core_parsing lines_hit=979 lines_found=1091 pct=89.73 suite=00:06 +98
core_embedding lines_hit=1250 lines_found=1458 pct=85.73 suite=00:11 +93

gate_new_packages_min=85
gate_core_parsing_min=80
gate_core_embedding_min=85
core_parsing_week2_snapshot=949/1043=90.99 on that week's 1043 instrumented
 lines
core_embedding_matches_week3_snapshot=1250/1458=85.73
# 2026-10-26: core_storage 分母 573 -> 666, 来自新增的 ScoredCandidate/rowI
dOf 与三阶段向量查询。
# 新增 10 条用例后命中 574 行，回到 85% 中枢之上。
```


新增 10 条用例后命中 374 行，回到 85% 门槛之上。
core_pipeline 93.96 -> 91.28：产物存在性用例归入 integration 后少覆盖 8 行，属下降但仍在门槛之上。

W4SHOT-H1

```
PS E:\Project\accessible-multimodal-retrieval> Get-Content -LiteralP
ath reports\week4\logs\storage_equivalence.txt
comparison=chroma_top5_vs_dart_sqlite_top5
slots_per_query=5
queries=25
positions=125
agreed=125
agreement=1.000000
gate=0.96
max_abs_score_delta_same_rank=0.00000039
max_abs_score_delta_same_chunk=0.00000039
tie_break=score_desc,relativePath_asc,page_asc_null_last,startOffset
_asc
query_vectors=exported_vectorB64
passed=1
```

W4SHOT-CAP

```
PS E:\Project\accessible-multimodal-retrieval> Get-Content -LiteralP
ath reports\week4\logs\storage_capacity.txt
sqlite_version=3.53.4
journal_mode=wal
n=1500 write_ms=95 query_ms_median=10 query_ms_min=8 query_ms_max=18
n=10000 write_ms=444 query_ms_median=44 query_ms_min=41 query_ms_max
=50
explain_filter=extension_in_sql
explain=3|0|60|SEARCH chunk USING INDEX idx_chunk_space (space=? AND
 dim=?)
write_limit_ms=600
query_limit_ms=60
```

```
accessible-multimodal-retrieval on  main [✗+?]
❯ Get-Content -LiteralPath reports\week4\logs\chroma_baseline.txt
chroma_vs_numpy_positions=125
chroma_vs_numpy_agreed=125
chroma_vs_numpy_agreement=1.000000
score=1-distance
hnsw_space=cosine
chromadb_version=1.5.9
queries=25
hit@5=0.880000 [0.700442, 0.958332] (22/25)
mrr@10=0.795714
metric_layer=storage_baseline
note=query vectors are the exported question embeddings; this is the Chroma control, not the product retriever
```

```
accessible-multimodal-retrieval on  main [✗+?]
❯ Get-Content -LiteralPath reports\week4\logs\e2e_build.txt
progress 33/50 E:\Project\accessible-multimodal-retrieval\datasets/coco/val_sample/images
progress 34/50 E:\Project\accessible-multimodal-retrieval\datasets/coco/val_sample/images
progress 35/50 E:\Project\accessible-multimodal-retrieval\datasets/coco/val_sample/images
progress 36/50 E:\Project\accessible-multimodal-retrieval\datasets/coco/val_sample/images
progress 37/50 E:\Project\accessible-multimodal-retrieval\datasets/coco/val_sample/images
progress 38/50 E:\Project\accessible-multimodal-retrieval\datasets/coco/val_sample/images
progress 39/50 E:\Project\accessible-multimodal-retrieval\datasets/coco/val_sample/images
progress 40/50 E:\Project\accessible-multimodal-retrieval\datasets/coco/val_sample/images
progress 41/50 E:\Project\accessible-multimodal-retrieval\datasets/coco/val_sample/images
progress 42/50 E:\Project\accessible-multimodal-retrieval\datasets/coco/val_sample/images
progress 43/50 E:\Project\accessible-multimodal-retrieval\datasets/coco/val_sample/images
progress 44/50 E:\Project\accessible-multimodal-retrieval\datasets/coco/val_sample/images
progress 45/50 E:\Project\accessible-multimodal-retrieval\datasets/coco/val_sample/images
progress 46/50 E:\Project\accessible-multimodal-retrieval\datasets/coco/val_sample/images
progress 47/50 E:\Project\accessible-multimodal-retrieval\datasets/coco/val_sample/images
progress 48/50 E:\Project\accessible-multimodal-retrieval\datasets/coco/val_sample/images
progress 49/50 E:\Project\accessible-multimodal-retrieval\datasets/coco/val_sample/images
progress 50/50 E:\Project\accessible-multimodal-retrieval\datasets/coco/val_sample/images
path=E:\Project\accessible-multimodal-retrieval\datasets/coco/val_sample/images
key_prefix=datasets/coco/val_sample/images
ingest_total=50 ingest_ok=50 ingest_partial=0 ingest_skipped=0 ingest_failed=0 ingest_cancelled=false ingest_ms=1418
documents=50 written=50 reused=0 failed=0 cancelled=false chunks=50 index_ms=7048
stats_chunks=147 stats_documents=53 stats_document_text=96 stats_clip_shared=51 stats_keyword_terms=0
index_ms_total=19093
```

```
accessible-multimodal-retrieval on  main [✗+?]
❯ dart run packages/core_pipeline/bin/index_probe.dart --db reports/week4/.index/evidence.sqlite --path datasets/samples --path datasets/coco/val_sample/images
progress 33/50 E:\Project\accessible-multimodal-retrieval\datasets/coco/val_sample/images
progress 34/50 E:\Project\accessible-multimodal-retrieval\datasets/coco/val_sample/images
progress 35/50 E:\Project\accessible-multimodal-retrieval\datasets/coco/val_sample/images
progress 36/50 E:\Project\accessible-multimodal-retrieval\datasets/coco/val_sample/images
progress 37/50 E:\Project\accessible-multimodal-retrieval\datasets/coco/val_sample/images
progress 38/50 E:\Project\accessible-multimodal-retrieval\datasets/coco/val_sample/images
progress 39/50 E:\Project\accessible-multimodal-retrieval\datasets/coco/val_sample/images
progress 40/50 E:\Project\accessible-multimodal-retrieval\datasets/coco/val_sample/images
progress 41/50 E:\Project\accessible-multimodal-retrieval\datasets/coco/val_sample/images
progress 42/50 E:\Project\accessible-multimodal-retrieval\datasets/coco/val_sample/images
progress 43/50 E:\Project\accessible-multimodal-retrieval\datasets/coco/val_sample/images
progress 44/50 E:\Project\accessible-multimodal-retrieval\datasets/coco/val_sample/images
progress 45/50 E:\Project\accessible-multimodal-retrieval\datasets/coco/val_sample/images
progress 46/50 E:\Project\accessible-multimodal-retrieval\datasets/coco/val_sample/images
progress 47/50 E:\Project\accessible-multimodal-retrieval\datasets/coco/val_sample/images
progress 48/50 E:\Project\accessible-multimodal-retrieval\datasets/coco/val_sample/images
progress 49/50 E:\Project\accessible-multimodal-retrieval\datasets/coco/val_sample/images
progress 50/50 E:\Project\accessible-multimodal-retrieval\datasets/coco/val_sample/images
path=E:\Project\accessible-multimodal-retrieval\datasets/coco/val_sample/images
key_prefix=datasets/coco/val_sample/images
ingest_total=50 ingest_ok=50 ingest_partial=0 ingest_skipped=0 ingest_failed=0 ingest_cancelled=false ingest_ms=1298
documents=50 written=0 reused=50 failed=0 cancelled=false chunks=0 index_ms=1316
stats_chunks=147 stats_documents=53 stats_document_text=96 stats_clip_shared=51 stats_keyword_terms=0
index_ms_total=7465
```

```
accessible-multimodal-retrieval on  main [✗+?] took 23s
❯ $inc = Join-Path $env:TEMP 'w4-shot-incremental'
· if (Test-Path $inc) { Remove-Item -LiteralPath $inc -Recurse -Force }
· New-Item -ItemType Directory -Path $inc | Out-Null
· Copy-Item docs\architecture\adr\0001-parsing-runtime-dart-package.md (Join-Path $inc 'a.md') …
progress 0/3 C:\Users\11758\AppData\Local\Temp\w4-shot-incremental
progress 1/3 C:\Users\11758\AppData\Local\Temp\w4-shot-incremental
progress 2/3 C:\Users\11758\AppData\Local\Temp\w4-shot-incremental
progress 3/3 C:\Users\11758\AppData\Local\Temp\w4-shot-incremental
path=C:\Users\11758\AppData\Local\Temp\w4-shot-incremental
key_prefix=C:/Users/11758/AppData/Local/Temp/w4-shot-incremental
ingest_total=3 ingest_ok=3 ingest_partial=0 ingest_skipped=0 ingest_failed=0 ingest_cancelled=false ingest_ms=80
documents=3 written=3 reused=0 failed=0 cancelled=false chunks=393 index_ms=17382
stats_chunks=393 stats_documents=3 stats_document_text=393 stats_clip_shared=0 stats_keyword_terms=3854
index_ms_total=17382
db=C:\Users\11758\AppData\Local\Temp\w4-shot-incremental\inc.sqlite
progress 0/3 C:\Users\11758\AppData\Local\Temp\w4-shot-incremental
progress 1/3 C:\Users\11758\AppData\Local\Temp\w4-shot-incremental
progress 2/3 C:\Users\11758\AppData\Local\Temp\w4-shot-incremental
progress 3/3 C:\Users\11758\AppData\Local\Temp\w4-shot-incremental
path=C:\Users\11758\AppData\Local\Temp\w4-shot-incremental
key_prefix=C:/Users/11758/AppData/Local/Temp/w4-shot-incremental
ingest_total=3 ingest_ok=3 ingest_partial=0 ingest_skipped=0 ingest_failed=0 ingest_cancelled=false ingest_ms=41
documents=3 written=1 reused=2 failed=0 cancelled=false chunks=85 index_ms=5158
stats_chunks=394 stats_documents=3 stats_document_text=394 stats_clip_shared=0 stats_keyword_terms=3871
index_ms_total=5158
```

```
accessible-multimodal-retrieval on  main [✗?] took 38s
❯ dart run packages/core_pipeline/bin/search_probe.dart --db reports/week4/.index/evidence.sqlite --top 10 --query "工具靠什么判断文件类型？"
query=工具靠什么判断文件类型？
arm=A5 expand=true top=10 encode_ms=2237 store_ms=35 fuse_ms=3
rank=1 score=0.008197 key=datasets/samples/sample.pdf file=sample.pdf page=1 start=423 end=517 channels=documentText modality=text
snippet=The Apache Tika toolkit detects and extracts metadata and text from over a thousand different\nfile types (such as PPT, XLS, and PDF). All of these file types can be parsed through a single\ninterface, making Tika useful for search engine indexing, content analysis, translation,
and
rank=2 score=0.008065 key=datasets/samples/sample.pdf file=sample.pdf page=2 start=430 end=524 channels=documentText modality=text
snippet=The Apache Tika toolkit detects and extracts metadata and text from over a thousand different\nfile types (such as PPT, XLS, and PDF). All of these file types can be parsed through a single\ninterface, making Tika useful for search engine indexing, content analysis, translation,
and
rank=3 score=0.007937 key=datasets/samples/sample.docx file=sample.docx page=null start=306 end=426 channels=documentText modality=text
snippet=Contributions and corrections are welcome.\nThe Apache Tika toolkit detects and extracts metadata and text from over a thousand different file types (such as PPT, X\nLS, and PDF).
rank=4 score=0.007813 key=datasets/samples/sample.pdf file=sample.pdf page=1 start=617 end=710 channels=documentText modality=text
snippet=interface, making Tika useful for search engine indexing, content analysis, translation, and\nmuch more. Tika identifies a file’s type from its bytes, so a missing or misleading extension\ndoes not change the result.
rank=5 score=0.007692 key=datasets/samples/sample.pdf file=sample.pdf page=2 start=626 end=719 channels=documentText modality=text
snippet=interface, making Tika useful for search engine indexing, content analysis, translation, and\nmuch more. Tika identifies a file’s type from its bytes, so a missing or misleading extension\ndoes not change the result.
rank=6 score=0.007576 key=datasets/samples/sample.docx file=sample.docx page=null start=440 end=560 channels=documentText modality=text
snippet=LS, and PDF).\nAll of these file types can be parsed through a single interface, making Tika useful for search engine indexing, content\nanalysis, translation, and much more.
rank=7 score=0.007463 key=datasets/samples/sample.pdf file=sample.pdf page=1 start=326 end=419 channels=documentText modality=text
snippet=but more work remains. Contributions and corrections are welcome.\nThe Apache Tika toolkit detects and extracts metadata and text from over a thousand different\nfile types (such as PPT, XLS, and PDF). All of these file types can be parsed through a single
rank=8 score=0.007353 key=datasets/samples/sample.pdf file=sample.pdf page=2 start=332 end=425 channels=documentText modality=text
snippet=but more work remains. Contributions and corrections are welcome.\nThe Apache Tika toolkit detects and extracts metadata and text from over a thousand different\nfile types (such as PPT, XLS, and PDF). All of these file types can be parsed through a single
rank=9 score=0.007246 key=datasets/samples/sample.docx file=sample.docx page=null start=599 end=709 channels=documentText modality=text
snippet=analysis, translation, and much more.\nTika identifies a file’s type from its bytes, so a missing or misleading extension does not change the result.\nWhere to start
rank=10 score=0.007143 key=datasets/samples/sample.docx file=sample.docx page=null start=725 end=826 channels=documentText modality=text
snippet=Where to start\nSupported Formats — what Tika parses, family by family, plus the generated parser-to-media-type list.\nUsing Tika — choose an integration method: Java API, command line, REST server, or gRPC.
```

```
accessible-multimodal-retrieval on  main [✗?] took 9s
❯ dart run packages/core_pipeline/bin/search_probe.dart --db reports/week4/.index/evidence.sqlite --top 10 --query "这份说明在能够被完全信任之前，还要求人做什么？"
query=这份说明在能够被完全信任之前，还要求人做什么？
arm=A5 expand=true top=10 encode_ms=2584 store_ms=37 fuse_ms=3
rank=1 score=0.008197 key=datasets/samples/sample.pdf file=sample.pdf page=1 start=160 end=253 channels=documentText modality=text
snippet=This reference guide was generated with the assistance of AI and requires human review\nbefore it can be fully trusted. This documentation serves as an example and a starting point,\nbut more work remains. Contributions and corrections are welcome.
rank=2 score=0.008065 key=datasets/samples/sample.pdf file=sample.pdf page=2 start=164 end=257 channels=documentText modality=text
snippet=This reference guide was generated with the assistance of AI and requires human review\nbefore it can be fully trusted. This documentation serves as an example and a starting point,\nbut more work remains. Contributions and corrections are welcome.
rank=3 score=0.007937 key=datasets/samples/sample.pdf file=sample.pdf page=1 start=1232 end=1314 channels=documentText modality=text
snippet=content you do not control.\n• Security — the trust model, and what to lock down before exposing tika-server or\ntika-grpc.
rank=4 score=0.007813 key=datasets/samples/sample.pdf file=sample.pdf page=2 start=1252 end=1334 channels=documentText modality=text
snippet=content you do not control.\n• Security — the trust model, and what to lock down before exposing tika-server or\ntika-grpc.
rank=5 score=0.007692 key=datasets/samples/sample.docx file=sample.docx page=null start=1160 end=1251 channels=documentText modality=text
snippet=Setting Limits and Robustness — bounding time, memory and output when parsing content you do not control.\nSecurity — the trust model, and what to lock down before exposing tika-server or tika-grpc.\nMigrating to 4.x — what changed since 3.x and what you have to update.
rank=6 score=0.007576 key=datasets/samples/sample.pdf file=sample.pdf page=1 start=2005 end=2053 channels=documentText modality=text
snippet=https://tika.apache.org/docs/4.0.x/advanced/robustness.html\nhttps://tika.apache.org/docs/4.0.x/security.html\nhttps://tika.apache.org/docs/4.0.x/migration-to-4x/index.html
rank=7 score=0.007463 key=datasets/samples/sample.pdf file=sample.pdf page=2 start=2041 end=2089 channels=documentText modality=text
snippet=https://tika.apache.org/docs/4.0.x/advanced/robustness.html\nhttps://tika.apache.org/docs/4.0.x/security.html\nhttps://tika.apache.org/docs/4.0.x/migration-to-4x/index.html
rank=8 score=0.007353 key=datasets/samples/sample.docx file=sample.docx page=null start=59 end=177 channels=documentText modality=text
snippet=Where to start\nThis reference guide was generated with the assistance of AI and requires human review before it can be fully trusted.\nThis documentation serves as an example and a starting point, but more work remains.
rank=9 score=0.007246 key=datasets/samples/sample.docx file=sample.docx page=null start=178 end=262 channels=documentText modality=text
snippet=This reference guide was generated with the assistance of AI and requires human review before it can be fully trusted.\nThis documentation serves as an example and a starting point, but more work remains.\nContributions and corrections are welcome.
rank=10 score=0.007143 key=datasets/samples/sample.pdf file=sample.pdf page=1 start=1881 end=1944 channels=documentText modality=text
snippet=https://tika.apache.org/docs/4.0.x/configuration/index.html\nhttps://tika.apache.org/docs/4.0.x/advanced/setting-limits.html\nhttps://tika.apache.org/docs/4.0.x/advanced/robustness.html
```

```
accessible-multimodal-retrieval on  main [✗+?] took 11s
❯ dart run packages/core_pipeline/bin/search_probe.dart --db reports/week4/.index/evidence.sqlite --top 5 --extension .pdf --query "工具靠什么判断文件类型？"
· dart run packages/core_pipeline/bin/search_probe.dart --db reports/week4/.index/evidence.sqlite --top 5 --path-prefix sample.pdf --query "工具靠什么判断文件类型？"
· dart run packages/core_pipeline/bin/search_probe.dart --db reports/week4/.index/evidence.sqlite --top 3 --modified-after 2020-01-01T00:00:00Z --query "工具靠什么判断文件类型？"
· dart run packages/core_pipeline/bin/search_probe.dart --db reports/week4/.index/evidence.sqlite --top 3 --modified-after 2099-01-01T00:00:00Z --query "工具靠什么判断文件类型？"
query=工具靠什么判断文件类型？
arm=A5 expand=true top=5 encode_ms=2379 store_ms=30 fuse_ms=3
rank=1 score=0.008197 key=datasets/samples/sample.pdf file=sample.pdf page=1 start=423 end=517 channels=documentText modality=text
snippet=The Apache Tika toolkit detects and extracts metadata and text from over a thousand different\nfile types (such as PPT, XLS, and PDF). All of these file types can be parsed through a single\ninterface, making Tika useful for search engine indexing, content analysis, translation,
and
rank=2 score=0.008065 key=datasets/samples/sample.pdf file=sample.pdf page=2 start=430 end=524 channels=documentText modality=text
snippet=The Apache Tika toolkit detects and extracts metadata and text from over a thousand different\nfile types (such as PPT, XLS, and PDF). All of these file types can be parsed through a single\ninterface, making Tika useful for search engine indexing, content analysis, translation,
and
rank=3 score=0.007937 key=datasets/samples/sample.pdf file=sample.pdf page=1 start=617 end=710 channels=documentText modality=text
snippet=interface, making Tika useful for search engine indexing, content analysis, translation, and\nmuch more. Tika identifies a file’s type from its bytes, so a missing or misleading extension\ndoes not change the result.
rank=4 score=0.007813 key=datasets/samples/sample.pdf file=sample.pdf page=2 start=626 end=719 channels=documentText modality=text
snippet=interface, making Tika useful for search engine indexing, content analysis, translation, and\nmuch more. Tika identifies a file’s type from its bytes, so a missing or misleading extension\ndoes not change the result.
rank=5 score=0.007692 key=datasets/samples/sample.pdf file=sample.pdf page=1 start=326 end=419 channels=documentText modality=text
snippet=but more work remains. Contributions and corrections are welcome.\nThe Apache Tika toolkit detects and extracts metadata and text from over a thousand different\nfile types (such as PPT, XLS, and PDF). All of these file types can be parsed through a single
query=工具靠什么判断文件类型？
arm=A5 expand=true top=5 encode_ms=2301 store_ms=34 fuse_ms=3
rank=1 score=0.008197 key=datasets/samples/sample.pdf file=sample.pdf page=1 start=423 end=517 channels=documentText modality=text
snippet=The Apache Tika toolkit detects and extracts metadata and text from over a thousand different\nfile types (such as PPT, XLS, and PDF). All of these file types can be parsed through a single\ninterface, making Tika useful for search engine indexing, content analysis, translation,
and
rank=2 score=0.008065 key=datasets/samples/sample.pdf file=sample.pdf page=2 start=430 end=524 channels=documentText modality=text
snippet=The Apache Tika toolkit detects and extracts metadata and text from over a thousand different\nfile types (such as PPT, XLS, and PDF). All of these file types can be parsed through a single\ninterface, making Tika useful for search engine indexing, content analysis, translation,
and
rank=3 score=0.007937 key=datasets/samples/sample.pdf file=sample.pdf page=1 start=617 end=710 channels=documentText modality=text
snippet=interface, making Tika useful for search engine indexing, content analysis, translation, and\nmuch more. Tika identifies a file’s type from its bytes, so a missing or misleading extension\ndoes not change the result.
rank=4 score=0.007813 key=datasets/samples/sample.pdf file=sample.pdf page=2 start=626 end=719 channels=documentText modality=text
snippet=interface, making Tika useful for search engine indexing, content analysis, translation, and\nmuch more. Tika identifies a file’s type from its bytes, so a missing or misleading extension\ndoes not change the result.
rank=5 score=0.007692 key=datasets/samples/sample.pdf file=sample.pdf page=1 start=326 end=419 channels=documentText modality=text
snippet=but more work remains. Contributions and corrections are welcome.\nThe Apache Tika toolkit detects and extracts metadata and text from over a thousand different\nfile types (such as PPT, XLS, and PDF). All of these file types can be parsed through a single
query=工具靠什么判断文件类型？
arm=A5 expand=true top=3 encode_ms=2138 store_ms=35 fuse_ms=3
rank=1 score=0.008197 key=datasets/samples/sample.pdf file=sample.pdf page=1 start=423 end=517 channels=documentText modality=text
snippet=The Apache Tika toolkit detects and extracts metadata and text from over a thousand different\nfile types (such as PPT, XLS, and PDF). All of these file types can be parsed through a single\ninterface, making Tika useful for search engine indexing, content analysis, translation,
and
rank=2 score=0.008065 key=datasets/samples/sample.pdf file=sample.pdf page=2 start=430 end=524 channels=documentText modality=text
snippet=The Apache Tika toolkit detects and extracts metadata and text from over a thousand different\nfile types (such as PPT, XLS, and PDF). All of these file types can be parsed through a single\ninterface, making Tika useful for search engine indexing, content analysis, translation,
and
rank=3 score=0.007937 key=datasets/samples/sample.docx file=sample.docx page=null start=306 end=426 channels=documentText modality=text
snippet=Contributions and corrections are welcome.\nThe Apache Tika toolkit detects and extracts metadata and text from over a thousand different file types (such as PPT, X\nLS, and PDF).
query=工具靠什么判断文件类型？
arm=A5 expand=true top=0 encode_ms=2408 store_ms=11 fuse_ms=0
hits=0
```

```
accessible-multimodal-retrieval on  main [✗+?] took 38s
❯ dart run packages/core_pipeline/bin/search_probe.dart --db reports/week4/.index/evidence.sqlite --top 5 --extension .jpg --query "哪张图里有人在厨房做披萨？"
query=哪张图里有人在厨房做披萨？
arm=A5 expand=true top=5 encode_ms=2575 store_ms=33 fuse_ms=3
rank=1 score=0.003279 key=datasets/coco/val_sample/images/000000403385.jpg file=000000403385.jpg page=null start=0 end=0 channels=clipShared modality=image
snippet=
rank=2 score=0.003226 key=datasets/coco/val_sample/images/000000296649.jpg file=000000296649.jpg page=null start=0 end=0 channels=clipShared modality=image
snippet=
rank=3 score=0.003175 key=datasets/coco/val_sample/images/000000463730.jpg file=000000463730.jpg page=null start=0 end=0 channels=clipShared modality=image
snippet=
rank=4 score=0.003125 key=datasets/coco/val_sample/images/000000006818.jpg file=000000006818.jpg page=null start=0 end=0 channels=clipShared modality=image
snippet=
rank=5 score=0.003077 key=datasets/coco/val_sample/images/000000329323.jpg file=000000329323.jpg page=null start=0 end=0 channels=clipShared modality=image
snippet=
```

```
accessible-multimodal-retrieval on  main [✗+?] took 9s
❯ dart run packages/core_pipeline/bin/search_probe.dart --db reports/week4/.index/evidence.sqlite --top 5 --arm A5 --query "工具靠什么判断文件类型？"
· dart run packages/core_pipeline/bin/search_probe.dart --db reports/week4/.index/evidence.sqlite --top 5 --arm A4 --query "工具靠什么判断文件类型？"
query=工具靠什么判断文件类型？
arm=A5 expand=true top=5 encode_ms=2614 store_ms=36 fuse_ms=3
rank=1 score=0.008197 key=datasets/samples/sample.pdf file=sample.pdf page=1 start=423 end=517 channels=documentText modality=text
snippet=The Apache Tika toolkit detects and extracts metadata and text from over a thousand different\nfile types (such as PPT, XLS, and PDF). All of these file types can be parsed through a single\ninterface, making Tika useful for search engine indexing, content analysis, translation,
and
rank=2 score=0.008065 key=datasets/samples/sample.pdf file=sample.pdf page=2 start=430 end=524 channels=documentText modality=text
snippet=The Apache Tika toolkit detects and extracts metadata and text from over a thousand different\nfile types (such as PPT, XLS, and PDF). All of these file types can be parsed through a single\ninterface, making Tika useful for search engine indexing, content analysis, translation,
and
rank=3 score=0.007937 key=datasets/samples/sample.docx file=sample.docx page=null start=306 end=426 channels=documentText modality=text
snippet=Contributions and corrections are welcome.\nThe Apache Tika toolkit detects and extracts metadata and text from over a thousand different file types (such as PPT, X\nLS, and PDF).
rank=4 score=0.007813 key=datasets/samples/sample.pdf file=sample.pdf page=1 start=617 end=710 channels=documentText modality=text
snippet=interface, making Tika useful for search engine indexing, content analysis, translation, and\nmuch more. Tika identifies a file’s type from its bytes, so a missing or misleading extension\ndoes not change the result.
rank=5 score=0.007692 key=datasets/samples/sample.pdf file=sample.pdf page=2 start=626 end=719 channels=documentText modality=text
snippet=interface, making Tika useful for search engine indexing, content analysis, translation, and\nmuch more. Tika identifies a file’s type from its bytes, so a missing or misleading extension\ndoes not change the result.
query=工具靠什么判断文件类型？
arm=A4 expand=false top=5 encode_ms=2806 store_ms=22 fuse_ms=3
rank=1 score=0.008197 key=datasets/samples/sample.pdf file=sample.pdf page=1 start=423 end=517 channels=documentText modality=text
snippet=file types (such as PPT, XLS, and PDF). All of these file types can be parsed through a single
rank=2 score=0.008065 key=datasets/samples/sample.pdf file=sample.pdf page=2 start=430 end=524 channels=documentText modality=text
snippet=file types (such as PPT, XLS, and PDF). All of these file types can be parsed through a single
rank=3 score=0.007937 key=datasets/samples/sample.docx file=sample.docx page=null start=306 end=426 channels=documentText modality=text
snippet=The Apache Tika toolkit detects and extracts metadata and text from over a thousand different file types (such as PPT, X
rank=4 score=0.007813 key=datasets/samples/sample.pdf file=sample.pdf page=1 start=617 end=710 channels=documentText modality=text
snippet=much more. Tika identifies a file’s type from its bytes, so a missing or misleading extension
rank=5 score=0.007692 key=datasets/samples/sample.pdf file=sample.pdf page=2 start=626 end=719 channels=documentText modality=text
snippet=much more. Tika identifies a file’s type from its bytes, so a missing or misleading extension
```

```
accessible-multimodal-retrieval on  main [✗+?] took 22s
❯ $cancel = Join-Path $env:TEMP 'w4-shot-cancel'
· if (Test-Path $cancel) { Remove-Item -LiteralPath $cancel -Recurse -Force }
· New-Item -ItemType Directory -Path $cancel | Out-Null
· Set-Content -LiteralPath (Join-Path $cancel 'a.txt') -Value '第一份文件保留。离线检索可以没有网络。' -Encoding utf8
· Set-Content -LiteralPath (Join-Path $cancel 'b.txt') -Value '第二份文件不应写入。键盘和显示器。' -Encoding utf8
· Set-Content -LiteralPath (Join-Path $cancel 'c.txt') -Value '第三份文件不应写入。向量数据库。' -Encoding utf8
· dart run packages/core_pipeline/bin/index_probe.dart --db (Join-Path $cancel 'cancel.sqlite') --path $cancel --cancel-after 1
· dart run packages/core_pipeline/bin/search_probe.dart --db (Join-Path $cancel 'cancel.sqlite') --top 5 --query "离线检索可以没有网络"
db=C:\Users\11758\AppData\Local\Temp\w4-shot-cancel\cancel.sqlite
progress 0/3 C:\Users\11758\AppData\Local\Temp\w4-shot-cancel
progress 1/3 C:\Users\11758\AppData\Local\Temp\w4-shot-cancel
path=C:\Users\11758\AppData\Local\Temp\w4-shot-cancel
key_prefix=C:/Users/11758/AppData/Local/Temp/w4-shot-cancel
ingest_total=3 ingest_ok=3 ingest_partial=0 ingest_skipped=0 ingest_failed=0 ingest_cancelled=false ingest_ms=99
documents=3 written=1 reused=0 failed=0 cancelled=true chunks=1 index_ms=3070
stats_chunks=1 stats_documents=1 stats_document_text=1 stats_clip_shared=0 stats_keyword_terms=15
index_ms_total=3070
query=离线检索可以没有网络
arm=A5 expand=true top=1 encode_ms=2411 store_ms=18 fuse_ms=2
rank=1 score=0.013115 key=C:/Users/11758/AppData/Local/Temp/w4-shot-cancel/a.txt file=a.txt page=null start=0 end=19 channels=documentText,keyword modality=text
snippet=第一份文件保留。离线检索可以没有网络。
```

```
accessible-multimodal-retrieval on  main [✗+?] took 32s
❯ $bad = Join-Path $env:TEMP 'w4-shot-bad'
· if (Test-Path $bad) { Remove-Item -LiteralPath $bad -Recurse -Force }
· New-Item -ItemType Directory -Path $bad | Out-Null
· Set-Content -LiteralPath (Join-Path $bad 'good.txt') -Value '完好的文本文件仍应入库。' -Encoding utf8
· Set-Content -LiteralPath (Join-Path $bad 'bad.docx') -Value 'this is not a docx archive' -Encoding utf8
· dart run packages/core_pipeline/bin/index_probe.dart --db (Join-Path $bad 'bad.sqlite') --path $bad
· dart run packages/core_pipeline/bin/search_probe.dart --db (Join-Path $bad 'bad.sqlite') --top 5 --query "完好的文本文件仍应入库"
db=C:\Users\11758\AppData\Local\Temp\w4-shot-bad\bad.sqlite
progress 0/1 C:\Users\11758\AppData\Local\Temp\w4-shot-bad
progress 1/1 C:\Users\11758\AppData\Local\Temp\w4-shot-bad
path=C:\Users\11758\AppData\Local\Temp\w4-shot-bad
key_prefix=C:/Users/11758/AppData/Local/Temp/w4-shot-bad
ingest_total=2 ingest_ok=1 ingest_partial=0 ingest_skipped=0 ingest_failed=1 ingest_cancelled=false ingest_ms=330
documents=1 written=1 reused=0 failed=0 cancelled=false chunks=1 index_ms=3054
stats_chunks=1 stats_documents=1 stats_document_text=1 stats_clip_shared=0 stats_keyword_terms=10
index_ms_total=3054
query=完好的文本文件仍应入库
arm=A5 expand=true top=1 encode_ms=2564 store_ms=47 fuse_ms=15
rank=1 score=0.013115 key=C:/Users/11758/AppData/Local/Temp/w4-shot-bad/good.txt file=good.txt page=null start=0 end=12 channels=documentText,keyword modality=text
snippet=完好的文本文件仍应入库。
```

```
accessible-multimodal-retrieval on  main [✗+?] took 30s
❯ $empty = Join-Path $env:TEMP 'w4-shot-empty'
· if (Test-Path $empty) { Remove-Item -LiteralPath $empty -Recurse -Force }
· New-Item -ItemType Directory -Path $empty | Out-Null
· dart run packages/core_pipeline/bin/index_probe.dart --db (Join-Path $empty 'empty.sqlite') --clear
· dart run packages/core_pipeline/bin/search_probe.dart --db (Join-Path $empty 'empty.sqlite') --top 5 --query "离线检索"
db=C:\Users\11758\AppData\Local\Temp\w4-shot-empty\empty.sqlite
clear_before_chunks=0 clear_after_chunks=0
query=离线检索
arm=A5 expand=true top=0 encode_ms=1637 store_ms=10 fuse_ms=0
hits=0
```

```
accessible-multimodal-retrieval on  main [?] took 18s
❯ $empty = Join-Path $env:TEMP 'w4-shot-empty'
· Set-Content -LiteralPath (Join-Path $empty 'note.txt') -Value '清库前的句子。离线检索。' -Encoding utf8
· dart run packages/core_pipeline/bin/index_probe.dart --db (Join-Path $empty 'empty.sqlite') --path $empty
· dart run packages/core_pipeline/bin/search_probe.dart --db (Join-Path $empty 'empty.sqlite') --top 5 --query "清库前的句子"
· dart run packages/core_pipeline/bin/index_probe.dart --db (Join-Path $empty 'empty.sqlite') --clear
· dart run packages/core_pipeline/bin/search_probe.dart --db (Join-Path $empty 'empty.sqlite') --top 5 --query "清库前的句子"
db=C:\Users\11758\AppData\Local\Temp\w4-shot-empty\empty.sqlite
progress 0/1 C:\Users\11758\AppData\Local\Temp\w4-shot-empty
progress 1/1 C:\Users\11758\AppData\Local\Temp\w4-shot-empty
path=C:\Users\11758\AppData\Local\Temp\w4-shot-empty
key_prefix=C:/Users/11758/AppData/Local/Temp/w4-shot-empty
ingest_total=1 ingest_ok=1 ingest_partial=0 ingest_skipped=0 ingest_failed=0 ingest_cancelled=false ingest_ms=86
documents=1 written=1 reused=0 failed=0 cancelled=false chunks=1 index_ms=2163
stats_chunks=1 stats_documents=1 stats_document_text=1 stats_clip_shared=0 stats_keyword_terms=8
index_ms_total=2163
query=清库前的句子
arm=A5 expand=true top=1 encode_ms=1898 store_ms=17 fuse_ms=1
rank=1 score=0.013115 key=C:/Users/11758/AppData/Local/Temp/w4-shot-empty/note.txt file=note.txt page=null start=0 end=12 channels=documentText,keyword modality=text
snippet=清库前的句子。离线检索。
db=C:\Users\11758\AppData\Local\Temp\w4-shot-empty\empty.sqlite
clear_before_chunks=1 clear_after_chunks=0
query=清库前的句子
arm=A5 expand=true top=0 encode_ms=1775 store_ms=8 fuse_ms=0
hits=0
```

```
accessible-multimodal-retrieval on  main [?]
❯ Select-String -LiteralPath reports\week4\logs\retrieval_bench_dev.txt -Pattern '^(parameters_changed_after_freeze|weights|k=|holdout_scored_runs|arm=)'

reports\week4\logs\retrieval_bench_dev.txt:1:parameters_changed_after_freeze=0
reports\week4\logs\retrieval_bench_dev.txt:2:weights=text:0.5,keyword:0.3,image:0.2
reports\week4\logs\retrieval_bench_dev.txt:3:k=60
reports\week4\logs\retrieval_bench_dev.txt:4:holdout_scored_runs=0
reports\week4\logs\retrieval_bench_dev.txt:20:arm=A0 split=dev n=15 multi_evidence=8 hit@5=0.000000 [0.000000, 0.203883] (0/15) mrr@10=0.000000 recall@5=0.000000 recall@10=0.000000 encode_ms_first=0 encode_ms_median_after_first=0.0 store_ms_median=0.0 fuse_ms_median=0.0
reports\week4\logs\retrieval_bench_dev.txt:36:arm=A1 split=dev n=15 multi_evidence=8 hit@5=0.666667 [0.417135, 0.848237] (10/15) mrr@10=0.622222 recall@5=0.633333 recall@10=0.666667 encode_ms_first=1732 encode_ms_median_after_first=79.0 store_ms_median=1.0 fuse_ms_median=0.0
reports\week4\logs\retrieval_bench_dev.txt:52:arm=A2 split=dev n=15 multi_evidence=8 hit@5=0.133333 [0.037361, 0.378820] (2/15) mrr@10=0.096296 recall@5=0.133333 recall@10=0.200000 encode_ms_first=437 encode_ms_median_after_first=52.5 store_ms_median=1.0 fuse_ms_median=0.0
reports\week4\logs\retrieval_bench_dev.txt:68:arm=A3 split=dev n=15 multi_evidence=8 hit@5=0.666667 [0.417135, 0.848237] (10/15) mrr@10=0.622222 recall@5=0.633333 recall@10=0.666667 encode_ms_first=37 encode_ms_median_after_first=83.5 store_ms_median=2.0 fuse_ms_median=0.0
reports\week4\logs\retrieval_bench_dev.txt:84:arm=A4 split=dev n=15 multi_evidence=8 hit@5=0.666667 [0.417135, 0.848237] (10/15) mrr@10=0.622222 recall@5=0.633333 recall@10=0.666667 encode_ms_first=67 encode_ms_median_after_first=78.5 store_ms_median=4.0 fuse_ms_median=0.0
reports\week4\logs\retrieval_bench_dev.txt:100:arm=A5 split=dev n=15 multi_evidence=8 hit@5=0.666667 [0.417135, 0.848237] (10/15) mrr@10=0.622222 recall@5=0.633333 recall@10=0.666667 encode_ms_first=59 encode_ms_median_after_first=127.0 store_ms_median=5.0 fuse_ms_median=0.0
```

```
accessible-multimodal-retrieval on  main [?]
❯ Get-Content -LiteralPath reports\week4\logs\retrieval_bench_holdout.txt
parameters_changed_after_freeze=0
weights=text:0.5,keyword:0.3,image:0.2
k=60
holdout_scored_runs=1
ev-16 split=holdout arm=A4 hit@5=1 rank=1 channels=documentText top5=[datasets/samples/sample.docx#pna@1, datasets/samples/sample.pdf#p1@2, datasets/samples/sample.pdf#p2@3, datasets/samples/sample.pdf#p1@4, datasets/samples/sample.pdf#p2@5]
ev-17 split=holdout arm=A4 hit@5=1 rank=1 channels=documentText top5=[datasets/samples/sample.pdf#p1@1, datasets/samples/sample.pdf#p2@2, datasets/samples/sample.docx#pna@3, datasets/samples/sample.docx#pna@4, datasets/samples/sample.pdf#p1@5]
ev-18 split=holdout arm=A4 hit@5=1 rank=1 channels=documentText top5=[datasets/samples/sample.docx#pna@1, datasets/samples/sample.pdf#p1@2, datasets/samples/sample.pdf#p2@3, datasets/samples/sample.pdf#p1@4, datasets/samples/sample.pdf#p2@5]
ev-19 split=holdout arm=A4 hit@5=1 rank=1 channels=documentText top5=[datasets/samples/sample.docx#pna@1, datasets/samples/sample.pdf#p1@2, datasets/samples/sample.pdf#p2@3, datasets/samples/sample.pdf#p1@4, datasets/samples/sample.pdf#p2@5]
ev-20 split=holdout arm=A4 hit@5=1 rank=1 channels=documentText top5=[datasets/samples/sample.docx#pna@1, datasets/samples/sample.pdf#p1@2, datasets/samples/sample.pdf#p2@3, datasets/samples/sample.docx#pna@4, datasets/samples/sample.pdf#p1@5]
ev-21 split=holdout arm=A4 hit@5=0 rank=0 channels= top5=[datasets/samples/sample.pdf#p1@1, datasets/samples/sample.pdf#p2@2, datasets/samples/sample.docx#pna@3, datasets/samples/sample.pdf#p1@4, datasets/samples/sample.pdf#p2@5]
ev-22 split=holdout arm=A4 hit@5=0 rank=0 channels= top5=[datasets/samples/sample.pdf#p1@1, datasets/samples/sample.pdf#p2@2, datasets/samples/sample.pdf#p1@3, datasets/samples/sample.pdf#p2@4, datasets/samples/sample.pdf#p1@5]
ev-23 split=holdout arm=A4 hit@5=0 rank=0 channels= top5=[datasets/samples/sample.pdf#p1@1, datasets/samples/sample.pdf#p2@2, datasets/samples/sample.pdf#p1@3, datasets/samples/sample.pdf#p2@4, datasets/samples/sample.pdf#p1@5]
ev-24 split=holdout arm=A4 hit@5=0 rank=0 channels= top5=[datasets/samples/sample.pdf#p1@1, datasets/samples/sample.pdf#p2@2, datasets/samples/sample.docx#pna@3, datasets/samples/sample.pdf#p1@4, datasets/samples/sample.pdf#p2@5]
ev-25 split=holdout arm=A4 hit@5=0 rank=0 channels= top5=[datasets/samples/sample.docx#pna@1, datasets/samples/sample.pdf#p1@2, datasets/samples/sample.pdf#p2@3, datasets/samples/sample.pdf#p1@4, datasets/samples/sample.pdf#p2@5]
arm=A4 split=holdout n=10 multi_evidence=4 hit@5=0.500000 [0.236593, 0.763407] (5/10) mrr@10=0.500000 recall@5=0.500000 recall@10=0.500000 encode_ms_first=2134 encode_ms_median_after_first=55.0 store_ms_median=21.5 fuse_ms_median=0.0
```

```
accessible-multimodal-retrieval on  main [✗+?]
❯ Get-Content -LiteralPath reports\week4\logs\offline_check.txt
# offline grep 2026-10-25T16:09:01.4371110+08:00
# roots: packages/core_storage/lib packages/core_retrieval/lib packages/core_pipeline/lib
# pattern: HttpClient|Socket|import .http|package:http
# files_scanned=40
# matches=0
# network_disabled=0
# note: grep gate only; the machine network was left enabled
```

检索精度基准报告

1. 口径

证据 `id` 等于 `documentKey`，或证据路径以 `/` 加相对路径结尾。Hit@5 是 Top-5 里出现任一相关文件。MRR@10 用第一次命中的名次。多文件题同时报 Recall。比例写成 `0.666667 [0.417135, 0.848237] (10/15)`。区间是 Wilson 95%。

第 2 节到第 8 节的权重是（文本，关键词，图像）= `(0.5, 0.3, 0.2)`，`k=60`。关掉一路不是改权重。两份冻结日志的第一行都是 `parameters_changed_after_freeze=0`。dev 先跑完全部六臂，之后 holdout 只跑 A4 一次。`retrieval_bench_dev.txt` 写着 `holdout_scored_runs=0`，`retrieval_bench_holdout.txt` 写着 `holdout_scored_runs=1`。2026-10-27 把图像权重改为 0.49，`parameters_changed_after_freeze=1`。那次测量在第 10 节，不改第 2 节。

2. 六臂

dev 15 条，其中 8 条有多份证据。holdout 10 条，其中 4 条有多份证据。A1、A3、A4、A5 在 15 条 dev 上的 Hit@5、名次和 Top-5 逐题相同。

臂	配置	dev Hit@5	dev MRR@10	dev Recall@5	dev Recall@10	holdout Hit@5	holdout MRR@10
A0	只开关键词，不扩展邻域	0.000000 [0.000000, 0.203883] (0/15)	0.000000	0.000000	0.000000	未跑	未跑
A1	只开 `documentText`	0.666667 [0.417135, 0.848237] (10/15)	0.622222	0.633333	0.666667	未跑	未跑

臂	配置	dev Hit@5	dev MRR@10	dev Recall@5	dev Recall@10	holdout Hit@5	holdout MRR@10
A2	只开 `clipShared`	0.133333 [0.037361, 0.378820] (2/15)	0.096296	0.133333	0.200000	未跑	未跑
A3	文本 + 关键词	0.666667 [0.417135, 0.848237] (10/15)	0.622222	0.633333	0.666667	未跑	未跑
A4	三路，不扩展邻域	0.666667 [0.417135, 0.848237] (10/15)	0.622222	0.633333	0.666667	0.500000 [0.236593, 0.763407] (5/10)	0.500000
A5	三路 + 邻域，产品默认	0.666667 [0.417135, 0.848237] (10/15)	0.622222	0.633333	0.666667	未跑	未跑

holdout 的 Recall@5 和 Recall@10 都是 0.500000。邻域不改名次，所以 A5 和 A4 的 Hit@5、MRR 相同。

命令：

```
engine\.venv\Scripts\python.exe engine\eval_retrieval.py --arms A0,A1,A2,A3,A4,A5 --split dev
engine\.venv\Scripts\python.exe engine\eval_retrieval.py --arms A4 --split holdout
```

3. 逐题

A4 是融合臂。A2 列只在 dev 上有数。名次 0 表示 Top-10 里没有相关文件。

题	划分	模态	A4 Hit@5	A4 名次	A2 名次	说明
ev-01	dev	文本	1	1	0	文件命中 `sample.pdf`
ev-02	dev	文本	1	1	0	文件命中 `sample.docx`
ev-03	dev	文本	1	1	0	文件命中 `sample.pdf`
ev-04	dev	文本	1	1	0	文件命中 `sample.docx`
ev-05	dev	文本	1	1	0	文件命中 `sample.pdf`
ev-06	dev	文本	1	1	0	文件命中 `sample.pdf`
ev-07	dev	文本	1	1	0	文件命中 `sample.docx`
ev-08	dev	文本	1	1	0	跨页题是两页重复排版，文件命中

题	划分	模态	A4 Hit@5	A4 名次	A2 名次	说明
ev-09	dev	元数据	1	3	0	创建者在 docProps，不在正文。Top-5 有 `sample.docx`
ev-10	dev	元数据	1	1	0	标题在 docProps。文件命中，不是属性跨度命中
ev-11	dev	图像	0	0	0	披萨图 `000000397133.jpg`。中文问句只开图像也不在 Top-5。英文证据句另见诊断日志，不改本行分数
ev-12	dev	图像	0	0	0	蓝自行车。只开图像也不在 Top-5
ev-13	dev	图像	0	0	1	破损淋浴墙。只开图像是第 1 名，融合后掉出
ev-14	dev	图像	0	0	9	客机。只开图像排第 9，未进 Hit@5
ev-15	dev	图像	0	0	3	`sample.png` 标题。只开图像是第 3 名，融合后掉出
ev-16	holdout	文本	1	1	未跑	文件命中
ev-17	holdout	文本	1	1	未跑	跨页重复排版，文件命中
ev-18	holdout	文本	1	1	未跑	文件命中
ev-19	holdout	文本	1	1	未跑	文件命中
ev-20	holdout	元数据	1	1	未跑	创建时间在 docProps。文件命中
ev-21	holdout	图像	0	0	未跑	融合 Top-5 全是文本文档
ev-22	holdout	图像	0	0	未跑	同上
ev-23	holdout	图像	0	0	未跑	同上

题	划分	模态	A4 Hit@5	A4 名次	A2 名次	说明
ev-24	holdout	图像	0	0	未跑	同上
ev-25	holdout	图像	0	0	未跑	同上

4. 耗时

建库是六臂共用的一次，不是每臂各建一次。`datasets/samples` 加 COCO 小样：第一次 `index_ms_total=19093`，147 块、53 个文档；第二次内容未改，`written=0`，`reused=53`，`index_ms_total=6873`。见 `reports/week4/logs/e2e_build.txt` 与 `reports/week4/logs/e2e_rebuild.txt`。

查询耗时来自 2026-10-26 的重跑，单位毫秒。A0 不编码，所以编码时间是 0。一个进程里的第一次编码含进程启动和模型加载，各次运行会差一截。dev 里 A1 的第一次编码是 1419 ms。holdout 是新进程，A4 第一次编码是 1757 ms。用来比较的是其后的中位。融合时间中位是 0.0，因为不到 1 ms。

臂	划分	首次编码	其后编码中位	存储查询中位	融合中位
A0	dev	0	0.0	0.0	0.0
A1	dev	1419	74.0	1.0	0.0
A2	dev	386	40.0	2.0	0.0
A3	dev	34	85.0	2.0	0.0
A4	dev	124	87.0	3.0	0.0
A5	dev	54	136.0	4.0	0.0
A4	holdout	1757	58.0	9.0	0.0

这组存储查询打在 147 块的证据库上。1 万块的向量查询是另一项测量，中位 44 ms，见第 6 节。

5. H2

门槛是：dev 上融合臂的 Hit@5 和 MRR@10 不低于任一单通道。

A4 等于 A1（10/15，MRR 0.622222），高于 A0（0/15）和 A2（2/15）。「不低于」成立。相对最好的单通道没有正增益。

原因有两条，都没有改权重。

样本库是英文。冻结集的问句是中文。`KeywordPlan.forQuery` 只要切出中文 bigram，就只走 bigram，不会落到正文上的 trigram。所以 A0 十五题都是空名单。这不是关键词通道关掉了。核对在第 8 节，不并进上面的分数表。A3 因此等于 A1。中文文档库不是这张表的语料；`e2e_docs.txt` 里同一套关键词代码有 `keyword_terms=14915`，能和文本一起命中 ADR 0004。

图像权重是 0.2。dev 上 A2 把 ev-13 排到第 1、ev-15 排到第 3，A4 的 Top-5 仍被 `sample.pdf` / `sample.docx` 占满，这两题变成未命中。ev-11 和 ev-12 的中文问句在只开图像时也不在 Top-5，不能记成融合单独造成的失败。ev-11 的英文证据句 `A man is in a kitchen making pizzas.` 只走 clipShared 时排第 1，分数 0.003279，即 `0.2 / (60 + 1)`。这条未命中是中文问句进不了 MobileCLIP 文本塔的词表（BPE，49408 个 token，汉字 token 为 0），不是图像向量对披萨语义没有区分度。图到图用的是库里的图像向量，第 1 名是原图自己，不检验「披萨」这个词。见 `logs/e2e07_clip_diagnosis.txt`。这不改上面的 Hit@5。图像权重后来改为 0.49，记在第 10 节。

6. 其他数字

Chroma 对照，`reports/week4/logs/chroma_baseline.txt`：chromadb 1.5.9，`hnsw:space=cosine`，用导出的问题向量，不是产品检索器。Hit@5 0.880000 [0.700442, 0.958332] (22/25)，MRR@10 0.795714。同一脚本里 Chroma 对 numpy 的 125 个位置一致率是 1.000000。

H1 以 `reports/week4/logs/storage_equivalence.txt` 为准：125 个位置同意 125 个，`agreement=1.000000`，`passed=1`。2026-10-26 用当前两份 jsonl 重跑了 `engine/compare_ranking.py`，档案数字与输入一致（此前那份 121/125 由更早的 Chroma 输出算出，已作废）。

1 万块产品写入与查询，`reports/week4/logs/storage_capacity.txt`：SQLite 3.53.4，WAL。写入 444 ms（门槛 600），查询中位 44 ms（最小 41，最大 50，门槛 60），两项都通过。`EXPLAIN` 解释的是生产扫描语句，走 `idx_chunk_space (space=? AND dim=?)`。门槛没有改。

7. 已知限制

dev 只有 15 条，holdout 只有 10 条。Hit@5 从 0.67 到 0.50 的差距落在很宽的 Wilson 区间里，不能读成稳定退步。没有「库里没有答案」的题。跨页题是重复排版。图像题一共 10 条。元数据题的文件能进 Top-5，属性原文不在正文块里。A0 为 0/15，是因为问句是中文、语料是英文：冻结运行时，有中文 bigram 的查询只走 bigram，这份库的 `keyword_term` 是 0 行，而 `chunk_fts` 上的英文 trigram 能命中。A1≡A3≡A4≡A5，六臂消融在这 25 条上区分不出关键词增益。融合没有超过文本单通道，权重保持冻结。ev-11 的中文问句进不了 MobileCLIP 文本塔的词表；英文证据句只走 clipShared 时该图排第 1。E2E-07 的冻结运行仍未通过。2026-10-27 的查询改写在第 9 节，图像权重改动在第 10 节，都不替换这些数字。

8. 关键词路径

这是 2026-10-27 查询改写之前的核对。库是 `reports/week4/.index/evidence.sqlite`。路径来自当时的 `KeywordPlan.forQuery`，命中来自 `search_probe --arm A0`。不改冻结集，也不把这两行写进 Hit@5。日志是 `logs/keyword_path_probe.txt`。改写之后的测量在第 9 节。

库内计数：`chunk_fts_rows=96`，`keyword_term_rows=0`。同一张 FTS5 trigram 表上，`Tika` 46 行，`metadata` 6 行，`xml` 0 行。

问句	KeywordPlan	A0
哪张图里有人在厨房做披萨?	bigram，11 个两字词	`hits=0`
Tika	trigram	`--top 5` 返回 5 条，片段里有 Tika。FTS 全量是 46，不是这 5 条就是全部

中文文档库是另一份索引，不进这 25 条。`logs/e2e_docs.txt` 里 `keyword_terms=14915`。问句「产品侧用系统 SQLite 保存向量」的第 1 名是 ADR 0004，通道是 `documentText,keyword`。

9. 查询改写

2026-10-27 做这次改写测量时，图像权重还是 0.2。同一天稍后改为 0.49，记在第 10 节。文档文本塔仍编码原中文问句。词表命中时，MobileCLIP 文本塔改为离线英文词；没有命中则仍用原问句。中文 bigram 仍走词表，另外把长度不少于 3 的拉丁串和这些英文词送进 trigram。整句中文不送进 trigram。同一块保留较高分。词表是仓库里的固定视觉词，不是证据句，也不是 jieba。日志是 `logs/query_rewrite_probe.txt`。

只开图像（A2）时，五条图像问句的金标文件都进了 Top-5。当时融合（A4）的 Top-5 里没有这些图像文件。图像块正文是空的，关键词打不中图像行。

问句	送给 CLIP 的字符串	A2 金标名次	冻结 A2
哪张图里有人在厨房做披萨?	person kitchen pizza	1，`000000397133.jpg`	未进 Top-5
哪张图里有一辆蓝色自行车停在路边?	blue bicycle parked roadside	1，`000000174482.jpg`	未进 Top-5
哪张图里的浴室淋浴墙是破的?	bathroom shower wall broken	2，`000000403385.jpg`	1
哪张图里有大型客机停在机场跑道上?	large airliner parked airport runway	1，`000000348881.jpg`	9
这张图片版面顶部的文档标题写的是什么?	picture layout top document title	1，`sample.png`	3

浴室那条从冻结的第 1 名变成第 2 名，仍在 Top-5 里。`sample.png` 排到第 1 只说明文件被召回，不说明模型读出了版面上的标题。

关键词：四条场景问句 `keyword_hits=0`。含「文档」「标题」的问句命中 `sample.pdf` / `sample.docx`，没有命中 `sample.png`。离线 的探针仍是空的，命中数 0。说明里的 `Tika` 没有词表命中，CLIP 字符串保持原问句，trigram 探针只有 `Tika`，前 5 条都是 `sample.docx`。权重仍是 0.2 时，A4 的五条 Top-5 都是这两份说明；标题问句的第 3 名和第 5 名片段是 `Apache Tika Documentation`，通道是 `documentText,keyword`。

10. 图像权重

文本权重仍是 0.5，关键词仍是 0.3，`k` 仍是 60。图像从 0.2 改为 0.49。`parameters_changed_after_freeze=1`。0.49 取自名次公式：大于 `0.5 * 62 / 64` 时，图像单独第 2 名进得了 Top-5；小于 0.5 时，文本单独第 3 名不会被同分路径序挤出。没有用 dev 标签搜索这个数。第 2 节和第 5 节的日志不改。新日志是 `logs/retrieval_bench_reweight_dev.txt`、`logs/retrieval_bench_reweight_holdout.txt`。holdout 这是第二次查看，不替换原来的 `retrieval_bench_holdout.txt`。0.900 的口径写在本节末段。

dev 上 A1 仍是 10/15，MRR 0.622222。A2 是 5/15，MRR 0.300000，五条都是图像题。A4 是 14/15，MRR 0.720741，高于 A1。A0 是 5/15，题号是 ev-04、ev-07、ev-08、ev-09、ev-10，图像题仍然是 0。ev-07 和 ev-09 的探针是词表里的 `document`，第 1 名是 `sample.docx` 的 `0–25`，正文是 `Apache Tika Documentation`，那是文档标题行，不是「说明正文」；同一问句的第 2 名 `178–262` 才是正文里的 `This documentation serves as an example and a starting point`。按文件算 Hit@5 记 1 成立，但 ev-07 与 ev-09 的首位命中来自标题行，写报告时不能把它当成正文命中。ev-10 另有探针 `text`，打在正文的 `metadata and text` 上；探针 `title` 在库里 0 命中。ev-04 的探针是问句里的 `Pipes`，第 1 名是 `Pipes — process-isolated` 那一行。`Pipes` 共 5 块：三块是正文里的 `Pipes — process-isolated, fault-tolerant extraction at scale`（docx 一块、pdf 两页各一块），两块是 pdf 页脚链到 `pipes/index.html` 的 URL。首位是正文，后面有页脚。ev-08 的探针是问句里的 `commit`，命中 `Built from commit: 47c1a0335`，3 块。`Pipes` 和 `commit` 不是词表。ev-01 到 ev-10 的 A4 仍全部命中，ev-09 从冻结时的第 3 名变成第 2 名。

冻结集的文本和元数据题是 15 条：dev 10 条、holdout 5 条。12 是 dev 这 10 条、50 个 Top-5 位置里的图像行，落在其中 6 条上，每条 2 行。冻结 A4 的这 50 个位置没有图像行。权重 0.49 之后是 12 行：ev-01、ev-02、ev-03、ev-05、ev-06 的第 3 名和第 5 名是图像，ev-08 的第 4 名和第 5 名是图像。ev-04、ev-07、ev-09、ev-10 的 Top-5 仍全是说明文件。金标文件仍在 Top-5 里，Hit@5 没有下降。图像单独第 1 名是 `0.49/61`，介于文本单独第 2 名 `0.5/62` 和第 3 名 `0.5/63` 之间，所以只走文本的块会把第 3 名让给图像。单测把前五名锁成文本第 1、文本第 2、图像第 1、文本第 3、图像第 2。ev-02 没有词表命中，Top-5 里仍有图像行。holdout 那 5 条里，ev-16、ev-18、ev-19 另有 6 行，不并进这 12，也不拿去调权重。

题	A4 名次	Top-5 里的金标
ev-11 披萨	3	`000000397133.jpg`
ev-12 蓝自行车	3	`000000174482.jpg`
ev-13 破损淋浴墙	5	金标 `000000403385.jpg` 在第 5 名。第 3 名是另一张图 `000000006818.jpg`
ev-14 客机	3	`000000348881.jpg`
ev-15 版面标题	9	未进 Top-5。Top-5 是 `sample.docx` / `sample.pdf`，通道含 keyword

ev-13 的 Top-5 里有两张图。图像行名额留 2 张时，第 3 名和第 5 名都在，这一题不变。名额留 1 张时留下第 3 名 `000000006818.jpg`，金标 `000000403385.jpg` 出 Top-5。

词表有 128 个中文键、117 个不同的英文词。其中 30 个英文字符串与 COCO 80 类的类别名相同。`orange` 在词表里是颜色，与 COCO 的水果类同名。`sofa`、`phone`、`ball`、`table` 和 COCO 的 `couch`、`cell phone`、`sports ball`、`dining table` 不是同一字符串。10 条图像题都命中词表，holdout 也在内。dev 上是「人 / 厨房 / 披萨」「蓝色 / 自行车 / 停在 / 路边」「浴室 / 淋浴 / 墙 / 破的」「大型 / 客机 / 停在 / 机场 / 跑道」「图片 / 版面 / 顶部 / 文档 / 标题」。holdout 上是「白色 / 房间 / 桶」「一群 / 山羊 / 人」「车 / 停在 / 楼」「长椅 / 水面 / 船」「图片 / 版面」。`山羊`、`桶`、`客机`、`路边`、`破的` 不是 COCO 类别名，它们出现在冻结问句里。15 条文本和元数据题里有 5 条命中：ev-01 的「人」，ev-07、ev-09、ev-10、ev-20 的「文档」，ev-10 还有「标题」「文字」。词表没有从 25 条英文 `evidenceText` 抄句子。冻结问句的措辞和词表键有重合，holdout 问句也在其中。改写之后的 holdout 命中，测的是这些已经覆盖的说法。这段只记录现有词表。

holdout 的 A4 是 9/10，MRR 0.620952。ev-16 到 ev-20 仍是第 1 名。ev-21 第 3，ev-22 第 3，ev-23 第 5，ev-24 第 5。ev-25 是第 7，Top-5 里没有 `sample.png`。版面上的字仍靠文本文件的正文和关键词，图像块正文是空的，0.49 盖不过两路都命中的说明块。产品检索的 holdout A4 已经查看 2 次：冻结日志是 5/10 = 0.500，这一次是改完查询和权重之后的重测。0.900 不能与 0.500 并列当成提升，也不再是无偏估计。冻结系统的那一次仍是 `retrieval_bench_holdout.txt`。本日志的 `holdout_scored_runs=1` 只计这一次进程。Chroma 对照排过这些查询向量，那是另一条存储路线，不计进这两次。